

Name: Meerab Asim
ID: F2023065216
Section: W-5

Assignment #4

Question #1

a) Where should the firewall be placed in the network?

The firewall should be positioned between the public internet and the organization's internal network. A web server should be placed in a DMZ (Demilitarized Zone) so customers can access it without exposing the internal network to external threats.

b) What type of traffic should be allowed?

The firewall should allow only legitimate and essential traffic. This includes secure customer access to the website through HTTPS, authorized remote connections such as VPNs, secure administrator logins, and other necessary communication between trusted systems.

c) What type of traffic should be blocked?

The firewall should block all unauthorized access attempts, malicious or suspicious traffic, and unnecessary ports or services. It should also prevent external users from directly accessing the organization's internal network.

d) What risks could arise if the firewall is configured incorrectly? Give any two examples.

- **Unauthorized access to systems:** If firewall rules are too permissive, attackers may gain access to internal systems and compromise sensitive information.
- **Exposure of the internal network:** An incorrect configuration may accidentally expose internal servers to the internet, making them vulnerable to cyberattacks.

e) Besides using a firewall, suggest two additional security measures that could help protect customer data and briefly explain their importance.

- **Antivirus software:** It helps detect, block, and remove malware, viruses, and other harmful software before they can infect systems or steal data.
- **Intrusion Detection System (IDS):** An IDS continuously monitors network and system activity, detects suspicious behavior, and alerts administrators to potential security threats so they can respond quickly.